

Hybrid Beamforming for Wideband Terahertz Massive MIMO Communications With Low-Resolution Phase Shifters and True-Time-Delay

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Abstract—Hybrid beamforming has attracted considerable interest in the development of next-generation wireless communications operating in the terahertz (THz) band. The short wavelength of THz wave has enabled thousands of antennas to be tightly packed inside a small area to provide sharp and accurate beamforming. Unfortunately, the beam squint effect, which changes the direction of beams as the frequency shifts, can only be circumvented by introducing true-time-delay (TTD) lines in the hybrid beamformers. However, most previous works neglected the resolution limitation of the precoding hardware and assumed phase shifters (PSs) and TTDs with infinite resolution. This paper proposes two quantized hybrid beamforming algorithms for the PS-only hybrid architecture and the TTD-assisted hybrid beamformer. We use a gradient descent approach with differentiable soft quantization techniques to overcome quantizer discontinuity. The RF precoder and the baseband precoder are alternately updated to maximize the similarity with the optimal precoder. Simulation results demonstrate the effectiveness of the proposed algorithms. The proposed 3-bit PS-only beamformer can achieve near-optimal performance and outperform other low-resolution precoding methods in a moderate squint scenario. The proposed low-resolution TTD-assisted architecture can effectively compensate severe squint effect and achieves near-optimal performance.

Index Terms—Terahertz, hybrid beamforming, true-time-delay, massive MIMO, low resolution.

I. INTRODUCTION

TERAHERTZ (THz) communications are promising candidates for the sixth generation (6G) communication networks [1], [2], [3], to provide a bandwidth of tens of gigahertz (GHz) due to the abundant unexplored frequency resources available in the THz spectrum (0.1-10THz). As new applications emerge, such as virtual reality (VR) [4] and

internet of things (IoT) [5], the demand for data bandwidth increases at an unprecedented rate [6]. Current communication systems operated at sub-6GHz or mmWave can hardly meet the required volume of traffic for these emerging applications. As a result, research on THz communications has recently gained increasing attention.

One primary concern of THz communications is its severe propagation loss due to the high radio attenuation at high frequencies [7]. Massive multiple-input-multiple-output (MIMO) transmission, which exploits antenna arrays with hundreds of antennas at the base station (BS) for a high beamforming gain, can provide enough signal enhancement and thus achieve reliable transmission [8]. Thanks to the short wavelength of THz waves, hundreds of antennas on a transceiver can be densely integrated within a relatively small area. Furthermore, recent development in micro-antennas [9] has made the implementation of a practical massive MIMO system possible in the THz band. However, traditional all-digital beamforming, where as many radio frequency (RF) chains as the number of antennas are required to convert the baseband signals to radio frequencies, becomes impractical due to the enormous hardware and power consumption cost.

Hybrid beamforming emerges as a low-complexity alternative [10], [11], [12], [13], [14], [15] to address the aforementioned issues, which decouples precoding into the baseband and RF precoding, where the RF precoding is implemented with a low-cost phase shifter (PS) network. Since most signal processing tasks are done by the PS network, the baseband processing as well as the number of RF chains can be significantly reduced. However, the gain of a phase shifter is not tunable, and the amount of phase shift is usually limited to several quantized angles [16], which introduces additional constraints on the precoder design. In addition, the frequency-flat nature of the phase shifters further reduces the beamforming performance in a frequency-selective wideband system.

The study of hybrid beamforming was first investigated in [11] for millimeter-wave (mmWave) systems. The design of the hybrid precoder was formulated as a sparse recovery problem. It was solved by an orthogonal matching pursuit (OMP) technique, but only limited to the single-subcarrier case. The work in [12] extended the result in [11] to

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the multi-carrier scheme, where the hybrid beamformer is designed by a greedy approach based on the Gram-Schmidt orthogonalization technique with a quantized RF codebook. In [14], a hybrid beamformer was derived by alternately updating the baseband and RF precoding coefficients to minimize the squared norm between the optimal all-digital precoder and the product of the RF and baseband precoders, based on the manifold optimization techniques.

In [13], the authors presented two heuristic hybrid beamforming algorithms for single-user MIMO (SU-MIMO) and multiuser multiple-input-single-output (MU-MISO) transmission in a frequency-selective environment while considering the quantization of PSs. Reference [17] proposed a codebook-based hybrid beamforming algorithm employing a partially-connected architecture for the wideband THz channel. A probabilistic approach was developed to maximize the average receive power when perfect channel state information (CSI) is unavailable. However, most of these studies often assume infinite-resolution PS, which is inconsistent with actual PS implementations [18], [19]. In light of this, several algorithms are proposed in [16], [20], and [21], specifically for hybrid precoders with low-resolution PSs. However, since small bandwidth is often assumed, these precoding methods often ignore the beam squint effect [22], an essential characteristic that must be considered in wideband transmission.

The spatial wideband effect [23] appears as the array aperture and signal bandwidth increase. In such a scenario, the propagation delay within an antenna array can easily exceed the signal sample time, rendering the phased-array approximation inaccurate. Consequently, the spatial wideband effect will result in the so-called beam squint effect [24] [25] in the frequency domain. This is illustrated in a traditional PS-based beamformer that the pointed angle seen by different frequencies will drift away from the intended physical direction. As a result, the beamforming gain at a specific frequency range deteriorates, lowering the overall system capacity.

Beam squint is traditionally mitigated by true-time-delay (TTD) lines in phased array radar system [26], [27], where large bandwidths are required. In such a system, PSs are replaced by TTDs to generate frequency-dependent phase responses to mitigate the beam squint effect. However, studies related to TTDs in hybrid beamforming systems remain largely unexplored. Recently, there have been some works focusing on the hybrid beamforming system with consideration of beam squint. In [28], squint-effect-aware CSI was projected on different subcarriers to the center frequency, and the RF beamformer was designed according to the projected CSI. The authors in [29] considered the beam squint effect in the channel model and adopted a hybrid beamforming scheme based on alternating minimization. A new hybrid beamformer architecture that connects two PSs to each antenna was proposed to alleviate beam squint with the help of the additional degree of freedom [30].

These methods all adopted the PS-only beamforming architecture, resulting in suboptimal precoding performance due to the inherent frequency-flat response of PSs. The authors in [31] introduced a TTD-assisted hybrid beamforming architecture, in which TTD lines are inserted between the RF precoder and

baseband precoder to compensate for the frequency-dependent phase shift. Also the work in [32] exploited similar architecture as [31] and proposed a closed-form derivation for the values of TTDs and PSs. However, both [31] and [32] ignored the finite resolution constraint of TTDs and PSs, and the result is valid only when a uniform linear array (ULA) is used. Finally, note that low-resolution TTDs are now silicon proven and can be integrated with RF circuits, such as mixers and amplifiers, without incurring too much area/power overhead [33], [34].

In this paper, we focus on hybrid beamforming design in a wideband massive MIMO-OFDM system in the presence of beam squint. We consider a three-dimensional (3D) THz channel, where uniform rectangular arrays (URA) are assumed at both the base station (BS) and mobile station (MS). We propose two effective hybrid beamforming algorithms for conventional fully-connected and TTD-assisted architecture respectively, considering the finite resolution of PSs and TTDs. The main contributions of this paper are listed as follows.

- We formulate the hybrid beamforming problem as a norm minimization problem to minimize the mean squared error (MSE) between the optimal all-digital beamformer and the overall hybrid beamformer. The RF precoder design problem is viewed as a single-layer neural network optimization problem, and gradient-based optimization techniques are used to update the values of PSs and TTDs. Then a simple least-squares approach derives the baseband precoder. We alternately update the RF and baseband precoders until the stopping criterion is met. A quantization-aware training (QAT) approach, called differentiable soft quantization (DSQ) [35], is incorporated into the proposed algorithm to derive the values of TTD and PS with low resolution, which yields better precoding performance than the naive quantization approach.
- We consider a wideband 3D spatial statistical channel model in the sub-THz band. Realistic spatial and temporal parameters are derived from existing measurement results [36] to formulate the propagation environment. URAs are assumed on both BS and MS to provide a 3D-steering capability. To the best of our knowledge, this is the first study of TDD-assisted hybrid beamforming associated with URA in a wideband 3D environment.
- Simulation results show that the proposed algorithm for conventional hybrid beamformer can outperform other low-resolution PS precoding schemes. The proposed TTD-assisted precoding scheme, on the other hand, can effectively alleviate the beam squint effect, yielding substantially higher spectral efficiency than its PS-only counterpart, while limiting the TTD resolution to a fraction of half-wavelength delay. Finally, we provide a comprehensive analysis of the impact of TTD number, arrangement, PS resolution, and TTD resolution on the achieved spectral efficiency.

The remainder of this paper is organized as follows. Section II introduces the system and channel models with beam squint. In Section III, we formulate the wideband hybrid beamforming problem. Section IV introduces the proposed hybrid beamforming algorithm for PS and TTD with limited

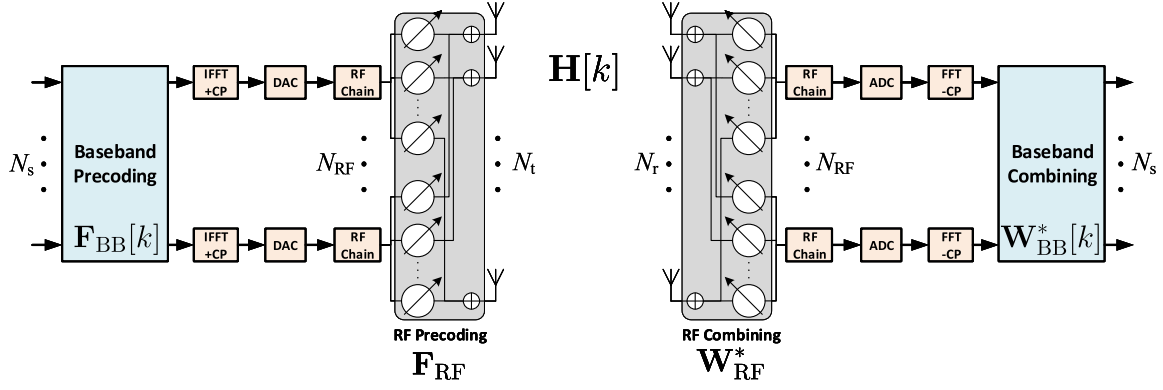


Fig. 1. A block diagram of the MIMO-OFDM transmit and receive system employing hybrid beamforming at both the Tx and Rx.

resolution. Section V provides the simulation results. Finally, we conclude the paper in Section VI.

Notations: Uppercase boldface, lowercase boldface and italic letters denote matrices, vectors and scalars, respectively. $(\cdot)^T$ and $(\cdot)^*$ denote transpose and conjugate transpose. $[\mathbf{A}]_{i,j}$, $[\mathbf{A}]_{:,j}$ and $[\mathbf{A}]_{i,:}$ represent the (i, j) th element, j th column and i th row of matrix $\mathbf{A}$. $[\mathbf{A}]_{:,S}$ denotes the matrix formed by concatenating the columns of $\mathbf{A}$ in the set S . $\|\mathbf{A}\|_F$, $|\mathbf{A}|$ and $\mathbf{A}^{-1}$ stand for the Frobenius norm, determinant and inverse of matrix $\mathbf{A}$, respectively. $\mathbf{I}_N$, $\mathbf{1}_{M \times N}$ and $\mathbf{0}_{M \times N}$ represent the $N \times N$ identity matrix, $M \times N$ all-one matrix and $M \times N$ all-zero matrix, respectively. $\angle(a)$, $|a|$ stand for the phase and modulus of complex number a , respectively. $\otimes$ and $\odot$ denote the Kronecker product and the Hadamard product, respectively. $\text{Re}(\cdot)$ and $\text{Im}(\cdot)$ denote the real part and the imaginary part, respectively. $\mathbb{E}[\cdot]$ denotes expectation. $\mathbb{R}$ and $\mathbb{C}$ denote the set of real numbers and the set of complex numbers. $j \triangleq \sqrt{-1}$ is the imaginary unit that is the square root of -1 .

II. SYSTEM AND CHANNEL MODELS

A. System Model

We consider a MIMO-OFDM system shown in Fig. 1, where N_t transmit antennas is installed in the base station (BS) to transmit N_s data streams to an mobile station (MS) with N_r receive antennas through N_{RF} RF chains. Note that the number of antennas is usually greater than the number of RF chains, and the size of the transmit antenna array is usually greater than the size of the receive array in a massive MIMO scenario, that is, $N_t > N_r \gg N_{RF} \geq N_s$. The data symbols $\mathbf{s}[k]$ on each subcarrier are first precoded by an $N_{RF} \times N_s$ precoder in the frequency domain, for $k = 1 \dots K$, where K is the total number of subcarriers. The precoded symbols are then converted to the time domain with N_{RF} K -point IFFTs and prepended with the cyclic prefix (CP). Finally, an $N_t \times N_{RF}$ RF precoder $\mathbf{F}_{RF}$ is applied to the time-domain symbols. The discrete-time transmit complex symbols $\mathbf{x}[k]$ at the k th subcarrier can be formulated as

$$\mathbf{x}[k] = \sqrt{\rho} \mathbf{F}_{RF} \mathbf{F}_{BB}[k] \mathbf{s}[k], \quad (1)$$

where $\mathbf{s}[k] \in \mathbb{C}^{N_s \times 1}$ is the complex transmit symbols, which satisfies $\mathbb{E}[\mathbf{s}[k] \mathbf{s}^*[k]] = \frac{1}{N_s} \mathbf{I}_{N_s}$. ρ is the average transmit power per subcarrier. $\mathbf{F}_{RF}$ and $\mathbf{F}_{BB}[k]$ are power constrained

by $\|\mathbf{F}_{RF} \mathbf{F}_{BB}[k]\|_F^2 = N_s$, for $k = 1 \dots K$. Besides, each entry of $\mathbf{F}_{RF}$ should meet the constant modulus and quantized phase constraints to represent limited-resolution phase shifters, given by

$$|[\mathbf{F}_{RF}]_{m,n}|^2 = 1, \quad (2)$$

$$\angle([\mathbf{F}_{RF}]_{m,n}) = \frac{2\pi q}{Q}, \quad (3)$$

where Q is the number of available phase shifts and $q = 0 \dots Q-1$.

At the MS side, the received symbols are first combined by the $N_r \times N_{RF}$ RF combiner $\mathbf{W}_{RF}$, which follows the same hardware constraint as $\mathbf{F}_{RF}$. After CP removal, the combined symbols are transformed into the frequency domain by the K -point FFT. Finally, each frequency-domain symbol is combined by an $N_{RF} \times N_s$ baseband combiner $\mathbf{W}_{BB}[k]$ for $k = 0 \dots K-1$. The received baseband symbols at the k th subcarrier can be expressed as

$$\mathbf{y}[k] = \sqrt{\rho} \mathbf{W}_{RF}^* \mathbf{W}_{BB}^*[k] \mathbf{H}[k] \mathbf{F}_{RF} \mathbf{F}_{BB}[k] \mathbf{s}[k] + \mathbf{W}_{RF}^* \mathbf{W}_{BB}^*[k] \mathbf{n}[k], \quad (4)$$

where $\mathbf{H}[k] \in \mathbb{C}^{N_r \times N_t}$ is the frequency-domain channel at the k th subcarrier. $\mathbf{n}[k] \in \mathbb{C}^{N_s \times 1}$ is the Gaussian noise vector with noise variance σ_n^2 .

B. Channel Model

Due to the limited scattering characteristics of THz channel, the THz channel can be modeled as the sum of multiple propagation paths with various angles and path gains, also known as *rays*. We consider a 3D geometric channel model [12], [17], [37], [38] and adjust it to include the beam squint effect, where several rays are statistically generated and clustered within multiple dominant directions. The wideband delay- d MIMO channel can be given by [12] and [39]

$$\mathbf{H}(d, f_k) = \gamma \sum_{i=1}^{N_{cl}} \sum_{\ell=1}^{L_i} \alpha_{i,\ell} p(dT_s - \tau_{i,\ell}) \times \mathbf{a}_r(\phi_{i,\ell}^r, \theta_{i,\ell}^r, f_k) \mathbf{a}_t^*(\phi_{i,\ell}^t, \theta_{i,\ell}^t, f_k), \quad (5)$$

where N_{cl} is the number of clusters. L_i denotes the number of rays within the i th cluster. $p(\cdot)$ refers to the raised-cosine

pulse shaping filter. T_s is the sampling period. $\alpha_{i,\ell}$ and $\tau_{i,\ell}$ are the path gain and the path delay of the ℓ th ray in the i th cluster, respectively. f_k is the center frequency of the k th subcarrier. γ is a normalization parameter such that $\mathbb{E} \left[\sum_{d=0}^{D-1} \|\mathbf{H}(d, f_k)\|_F^2 \right] = N_t N_r$, where D is the number of delay taps. $\phi_{i,\ell}^t, \theta_{i,\ell}^t, \phi_{i,\ell}^r$ and $\theta_{i,\ell}^r$ are the azimuth angle of departure (AoD), the elevation angle of departure (EoD), the azimuth angle of arrival (AoA) and the elevation angle of arrival (EoA), respectively. $\mathbf{a}_t^* \left(\phi_{i,\ell}^t, \theta_{i,\ell}^t, f_k \right)$ and $\mathbf{a}_r \left(\phi_{i,\ell}^r, \theta_{i,\ell}^r, f_k \right)$ are the beam-squint-adjusted frequency-dependent transmit and receive URA response vectors, respectively, and

$$\mathbf{a}(\phi, \theta, f) = \left[1 e^{j\pi \frac{f}{f_c} \cos \theta \sin \phi} \dots e^{j\pi \frac{f}{f_c} (N-1) \cos \theta \sin \phi} \right]^T \otimes \left[1 e^{j\pi \frac{f}{f_c} \sin \theta} \dots e^{j\pi \frac{f}{f_c} (M-1) \sin \theta} \right]^T, \quad (6)$$

for an $M \times N$ URA and f_c is the center frequency. Taking the discrete Fourier transform (DFT) of (5) along the delay d yields the discrete frequency-domain channel at the k th subcarrier as

$$\mathbf{H}[k] = \sum_{d=0}^{D-1} \mathbf{H}(d, f_k) e^{-j \frac{2\pi k}{K} d}. \quad (7)$$

C. Beam Squint and TTD-Assisted Beamforming

In a wideband system, the hybrid architecture based on the PS network, as shown in Fig. 1 suffers from severe performance loss due to the non-negligible beam squint effect. Given the fact that the phase shift associated with a particular delay τ at frequency f is given by $-j2\pi f\tau$. Traditional phased array systems approximate the phase shift by a frequency-invariant phase shift $-j2\pi f_c\tau$ for simple one-tap compensation, which can be implemented with simple hardware like a phase shifter. The approximation is almost accurate since $f \approx f_c$ in a narrowband scenario. However, PS is essentially a frequency-flat device, which cannot perfectly compensate for the frequency-varying phase shift. As the number of antennas and signal bandwidth in a THz system increase, the performance degradation caused by beam squint becomes more severe.

Considering a PS-based beamformer, whose antenna response is given by $\mathbf{f}(\phi, \theta) = \mathbf{a}_t(\phi_0, \theta_0, f_c)$ with a pointed direction at (ϕ_0, θ_0) . The normalized beamforming gain in the spatial direction (ϕ_0, θ_0) with frequency f can be formulated as

$$\begin{aligned} g(\phi, \theta, f) &= \frac{1}{MN} |\mathbf{a}_t^H(\phi, \theta, f) \mathbf{f}(\phi_0, \theta_0)| \\ &= \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N e^{-j\pi \frac{f}{f_c} ((n-1) \cos \theta \sin \phi + (m-1) \sin \theta)} \\ &\quad \times e^{j\pi ((n-1) \cos \theta_0 \sin \phi_0 + (m-1) \sin \theta_0)} \\ &= \frac{\sin \left(\frac{\pi N}{2} \left(\cos \theta_0 \sin \phi_0 - \frac{f}{f_c} \cos \theta \sin \phi \right) \right)}{N \sin \left(\frac{\pi}{2} \left(\cos \theta_0 \sin \phi_0 - \frac{f}{f_c} \cos \theta \sin \phi \right) \right)} \end{aligned}$$

$$\times \frac{\sin \left(\frac{\pi M}{2} \left(\sin \theta_0 - \frac{f}{f_c} \sin \theta \right) \right)}{M \sin \left(\frac{\pi}{2} \left(\sin \theta_0 - \frac{f}{f_c} \sin \theta \right) \right)}. \quad (8)$$

According to (8), it is obvious that $g(\phi_0, \theta_0, f) < g(\phi_0, \theta_0, f_c)$ when $f \neq f_c$. To put it shortly, at frequencies other than the center frequency, beamforming gain will degrade in the desired direction. As a result, the actual steered physical direction (ϕ', θ') at the frequency f can be formulated as

$$\theta' = \sin^{-1} \frac{f_c}{f} \sin \theta_0, \quad (9)$$

$$\phi' = \sin^{-1} \frac{\cos \theta_0 \sin \phi_0}{\sqrt{1 - \frac{f_c^2}{f^2} \sin^2 \theta_0}} \frac{f_c}{f}. \quad (10)$$

We further define the angle offset in azimuth as $\Delta_\phi = \phi' - \phi_0$ and the frequency offset ratio as $\frac{f}{f_c}$. Fig. 2(a) depicts the amount of angle offset as the pointed angle changes in various frequency offset ratios, where we can see that Δ_ϕ increases as the frequency offset ratio increases. The angle offset can be troublesome in a THz system with a massive number of antennas, where the beamwidth is usually extremely small. Therefore, a slight angle offset may result in a severe loss in the beamforming gain, as shown in Fig. 2(b). From Fig. 2(c), we can see that the loss in the beamforming gain becomes more significant when more antennas are used.

The beam squint effect is traditionally mitigated by replacing PSs with TTDs, e.g., in the phased-array radar applications [26], [27]. TTD can generate frequency-dependent phase shifts to perfectly compensate the beam squint effect. However, due to the associated increase in hardware cost, simply replacing all PSs with TTDs is impractical. In this paper, we consider a TTD-assisted architecture shown in Fig. 3(a), where several baseband or intermediate frequency (IF) TTDs are inserted between the fully-connected PS network and the baseband precoder. The architecture is first proposed in [31] to alleviate beam squint. We consider an $N_t = M \times N$ URA with several TTDs connected between the PS network and the baseband precoder. We assume that each TTD is connected to all PSs within a $U \times V$ subarray, as depicted in Fig. 3(b). Then we define the number of TTDs required to cover N_t PSs as N_{TTD} , whose value is given by $M/U \times N/V = N_t/UV$. Then, the frequency-selective RF precoder can be formulated as

$$\mathbf{F}_{\text{RF}}[k] = \mathbf{F}_{\text{PS}} \odot \mathbf{F}_{\text{TTD}}[k], \quad (11)$$

where $\mathbf{F}_{\text{PS}} \in \mathbb{C}^{N_t \times N_{\text{RF}}}$ is the precoding matrix formed by the PS network, which satisfies the constraints stated in (2) and (3). $\mathbf{F}_{\text{TTD}}[k] \in \mathbb{C}^{N_t \times N_{\text{RF}}}$ is the frequency response of the TTD delay lines applied to each targeted PS on the k th subcarrier. Specifically,

$$\begin{aligned} [\mathbf{F}_{\text{TTD}}[k]]_{\mathcal{P}_\ell} &= \exp \left(-j2\pi [\mathbf{T}]_{\ell,n} \bar{f}_k \right), \\ n &= 1 \dots N_{\text{RF}}, \quad \ell = 1 \dots N_{\text{TTD}}, \end{aligned} \quad (12)$$

where $\mathbf{T}$ is an $N_{\text{TTD}} \times N_{\text{RF}}$ matrix whose entries are the time delays of the TTDs. $\mathcal{P}_\ell$ is the set of PS indices, where the ℓ th TTD is connected to, in each RF chain, and $\bar{f}_k = f_k - f_c$ is the baseband frequency of the k th subcarrier. It is worth

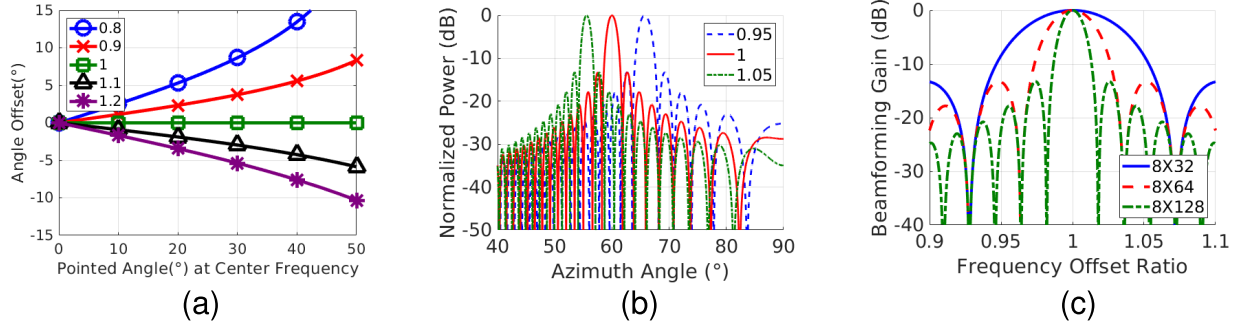


Fig. 2. Illustration of the beam squint effect. (a) The value of angle offset as a function of intended beamformed angle at different frequency offset ratio. (b) Illustration of beam pattern shifts with a 8×128 URA when $(\phi_0, \theta_0) = (60^\circ, 0^\circ)$. (c) Beamforming gain degradation as a function of frequency offset ratio with different antenna array size. (ϕ_0, θ_0) is set to $(60^\circ, 0^\circ)$.

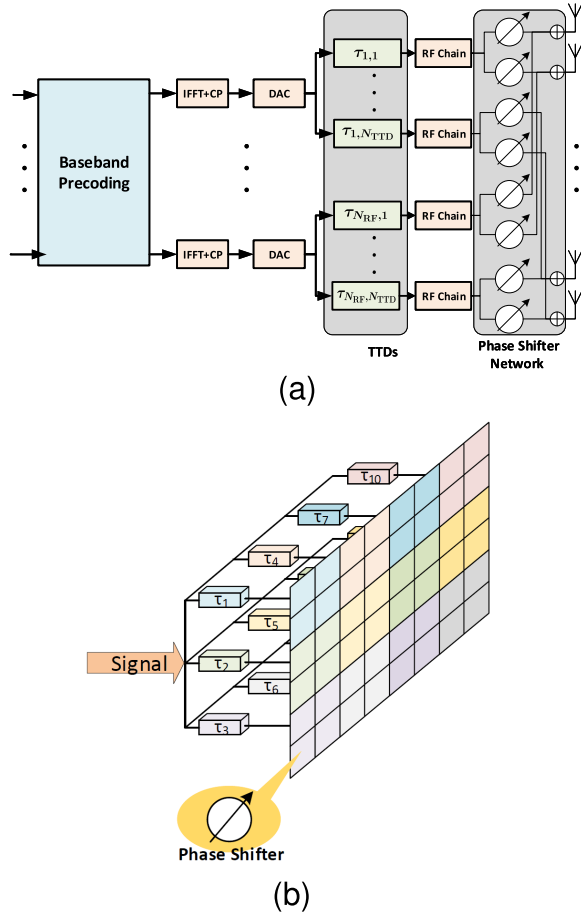


Fig. 3. TTD-assisted hybrid beamforming architecture. (a) Block diagram of the system. (b) Illustration of the TTD connection scheme.

mentioning that the insertion of TTDs enables $\mathbf{F}_{\text{RF}}[k]$ to have a frequency-selective response, allowing it to compensate for beam squint.

We consider baseband TTDs rather than the RF TTDs in [31] for two reasons. The first is that analog baseband TTDs typically have extremely low insertion loss, compact size, and wide delay range [40], making hardware implementation more cost-effective. The second reason is that the path-delay-induced phase shift can be intuitively decoupled as $-j2\pi f_k \tau = -j2\pi f_c \tau - j2\pi \bar{f}_k \tau$, where the first term is a

constant phase shift, which can be implemented with phase shifters. The second term is equivalent to the amount of phase shift incurred by the time delay τ at the baseband frequency $\bar{f}_k$, which can be exactly compensated by TTDs operating in the baseband frequency.

III. PROBLEM FORMULATION

Considering the hybrid transmit and receive system shown in Fig. 1, we can formulate the average spectral efficiency of the system as [41]

$$R = \frac{1}{K} \sum_{k=0}^{K-1} \log_2 \left(\left| \mathbf{I}_{N_s} + \frac{\rho}{N_s} \mathbf{R}_n^{-1}[k] \mathbf{H}_{\text{eff}}[k] \mathbf{H}_{\text{eff}}^*[k] \right| \right), \quad (13)$$

where $\mathbf{H}_{\text{eff}}[k] = \mathbf{W}_{\text{BB}}^*[k] \mathbf{W}_{\text{RF}}^* \mathbf{H}[k] \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]$ is the effective channel. $\mathbf{R}_n[k] = \sigma_n^2 \mathbf{W}_{\text{BB}}^*[k] \mathbf{W}_{\text{RF}}^* \mathbf{W}_{\text{RF}} \mathbf{W}_{\text{BB}}[k]$ is the noise covariance matrix after combining. If TTDs are used, $\mathbf{F}_{\text{RF}}$ should be replaced by frequency-variant $\mathbf{F}_{\text{RF}}[k]$ given in (11). In this paper, we mainly focus on the precoder design and assume that perfect synchronization, perfect CSI/equalization, and optimal demodulation and decoding are performed at the receiver side. By making this assumption, we can decouple the joint-precoding-combining optimization problem and only consider the average mutual information of the system, which is formulated as

$$\begin{aligned} & \mathcal{I} \left(\mathbf{F}_{\text{RF}}, \{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1} \right) \\ &= \frac{1}{K} \sum_{k=0}^{K-1} \log_2 \left| \mathbf{I} + \frac{\rho}{N_s \sigma_n^2} \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}[k] \mathbf{H}[k] \right. \\ & \quad \times \left. \mathbf{H}^*[k] \mathbf{F}_{\text{BB}}^*[k] \mathbf{F}_{\text{RF}}^* \right|. \end{aligned} \quad (14)$$

We seek to find the optimal set of RF and baseband precoders to maximize the mutual information given in (14), while satisfying the constraint of coefficient quantization and precoder power. The quantized PS hybrid beamforming problem is formally stated as follows.

$$\begin{aligned} & \mathbf{F}_{\text{RF}}^*, \{\mathbf{F}_{\text{BB}}^*[k]\}_{k=0}^{K-1} \\ &= \arg \max_{\mathbf{F}_{\text{RF}}, \{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1}} \mathcal{I} \left(\mathbf{F}_{\text{RF}}, \{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1} \right), \\ & \text{s.t. } \|\mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F^2 = N_s, \mathbf{F}_{\text{RF}} \text{ satisfies (2)(3)}. \end{aligned} \quad (15)$$

However, the complexity of directly optimizing the expression given in (15) is prohibitively high. It is shown in [11] that the given mutual information maximization problem is equivalent to minimizing the Euclidean distance between the optimal all-digital precoder and the overall hybrid precoder as follows.

$$\begin{aligned} & \mathbf{F}_{\text{RF}}^*, \{\mathbf{F}_{\text{BB}}^*[k]\}_{k=0}^{K-1} \\ &= \arg \min_{\mathbf{F}_{\text{RF}}, \{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1}} \frac{1}{K} \sum_{k=0}^{K-1} \|\mathbf{F}_{\text{opt}}[k] - \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F^2, \\ & \text{s.t. } \|\mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F^2 = N_s, \mathbf{F}_{\text{RF}} \text{ satisfies (2)(3)}, \end{aligned} \quad (16)$$

where $\mathbf{F}_{\text{opt}}[k] \in \mathbb{C}^{N_t \times N_s}$ is the capacity-achieving all-digital precoder at the k th subcarrier. We denote the singular value decomposition (SVD) of $\mathbf{H}[k]$ by $\mathbf{H}[k] = \mathbf{U}[k] \mathbf{\Sigma}[k] \mathbf{V}[k]$. Then $\mathbf{F}_{\text{opt}}[k]$ can be written as [41]

$$\mathbf{F}_{\text{opt}}[k] = [\mathbf{V}[k]]_{:,1:N_s} \mathbf{\Lambda}[k], \quad (17)$$

where $\mathbf{\Lambda}[k]$ is a $N_s \times N_s$ diagonal power allocation matrix at the k th subcarrier, which satisfies the principle of water-filling, given as

$$[\mathbf{\Lambda}[k]]_{i,i} = \sqrt{\left(\mu - \frac{N_s \sigma_n^2}{\rho [\mathbf{\Sigma}[k]]_{i,i}^2} \right)^+}, \text{ for } i = 1 \dots N_s, \quad (18)$$

where x^+ is defined as $\max(x, 0)$, and μ is the water level that satisfies the per-subcarrier power constraint as

$$\sum_{i=1}^{N_s} \left(\mu - \frac{N_s \sigma_n^2}{\rho [\mathbf{\Sigma}[k]]_{i,i}^2} \right)^+ = N_s. \quad (19)$$

For the TTD-assisted architecture shown in Fig. 3(a), the problem in (16) can be re-formulated as the following quantized-PS-TTD hybrid beamforming problem.

$$\begin{aligned} & \mathbf{T}^*, \mathbf{F}_{\text{PS}}^*, \{\mathbf{F}_{\text{BB}}^*[k]\}_{k=0}^{K-1} \\ &= \arg \min_{\mathbf{T}, \mathbf{F}_{\text{PS}}, \{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1}} \left\{ \frac{1}{K} \sum_{k=0}^{K-1} \|\mathbf{F}_{\text{opt}}[k] - (\mathbf{F}_{\text{PS}} \odot \mathbf{F}_{\text{TTD}}[k]) \mathbf{F}_{\text{BB}}[k]\|_F^2 \right\}, \\ & \text{s.t. } \|(\mathbf{F}_{\text{PS}} \odot \mathbf{F}_{\text{TTD}}[k]) \mathbf{F}_{\text{BB}}[k]\|_F^2 = N_s, \\ & \quad |\mathbf{F}_{\text{PS}}[m,n]| = 1, \angle(\mathbf{F}_{\text{PS}}[m,n]) = \frac{2\pi q}{Q}, [\mathbf{T}]_{m,n} = i\Delta_\tau, \end{aligned} \quad (20)$$

where Δ_τ is the minimum interval of TTD, and i is a non-zero integer. It is worth noting that the limited resolution constraints given in (16) and (20) make the overall hybrid beamforming design a non-convex optimization problem. A naive approach is to apply existing infinite resolution hybrid beamforming algorithms [11], [12], [14] and then directly quantize the obtained hybrid beamforming coefficients. However, direct quantization will result in severe performance loss. In the following section, we propose two algorithms to effectively solve the limited-resolution hybrid beamforming design problems. One is for the PS-only architecture as described in (16), and the other is for the TTD-assisted architecture as defined in (20).

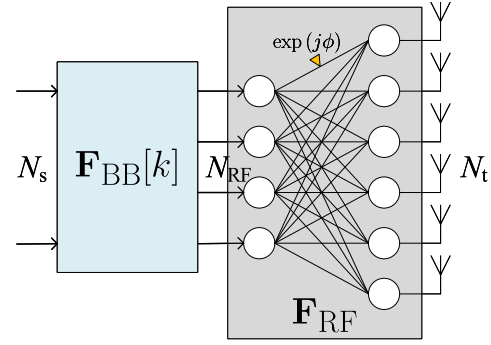


Fig. 4. The RF beamformer can be viewed as a single-layer fully-connected neural-network with complex constant modulus weights.

IV. HYBRID BEAMFORMING WITH LOW-RESOLUTION PSS AND TTDs

This section details the two proposed algorithms to solve the quantized hybrid beamforming problems with and without TTDs. The proposed algorithms are based on the idea of modeling the RF beamformer as a single-layer fully-connected neural network (NN) from a machine learning point of view, as illustrated in Fig. 4. The phase shifters are analogous to the NN weights but with a constant modulus and a quantized phase shift. The beamforming design process is similar to how gradient descent works in machine learning, where a collection of weights are iteratively updated to minimize a specific cost function. We employ a gradient-descent-based approach to update the phase shifts and the TTD values that comprise the RF precoder to minimize the MSE between the all-digital beamformer and the hybrid beamformer. However, the discontinuity of the quantizer leads to zero or infinite derivatives that prevent the weights from effective updating. This issue is resolved by incorporating a differentiable quantization function (DSQ) [35] in the proposed algorithms.

A. Limited-Resolution Hybrid Beamforming With PS-Only Architecture

This subsection describes the proposed hybrid beamforming design algorithm for PS-only architecture. The idea is then extended to the next subsection for TTD-assisted architecture. We start by first defining the phase-quantized loss function, given as

$$\mathcal{L} = \frac{1}{K} \sum_{k=0}^{K-1} \|\mathbf{F}_{\text{opt}}[k] - \bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F^2, \quad (21)$$

where $\bar{\mathbf{F}}_{\text{RF}}$ denotes the phase-quantized RF precoder, which is computed as $\bar{\mathbf{F}}_{\text{RF}} = \exp(j\bar{\Phi})$, and $\bar{\Phi}$ is the quantized phase shift, defined as $\bar{\Phi} = \left\lfloor \frac{\Phi}{\Delta_\phi} \right\rfloor \cdot \Delta_\phi$, where $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer, $\Phi \in \mathbb{R}^{N_t \times N_{\text{RF}}}$ is the unquantized phase shifts, and Δ_ϕ is the phase shift resolution, defined as $\frac{2\pi}{Q}$. We seek to find the gradient of $\mathcal{L}$ with respect to the phase shift Φ , denoted as $\nabla_\Phi \mathcal{L}$, where $[\nabla_\Phi \mathcal{L}]_{m,n} \triangleq \frac{\partial \mathcal{L}}{\partial [\Phi]_{m,n}}$. First, we separate the real part and the imaginary part of $\mathcal{L}$ and

express the gradient as

$$\begin{aligned}\nabla_{\Phi} \mathcal{L} &= \frac{1}{K} \sum_{k=0}^{K-1} \nabla_{\Phi} \mathcal{L}_k \\ &= \frac{1}{K} \sum_{k=0}^{K-1} \nabla_{\Phi} \left\{ \left\| \text{Re}(\mathbf{F}_{\text{opt}}[k]) - \text{Re}(\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]) \right\|_F^2 \right. \\ &\quad \left. + \left\| \text{Im}(\mathbf{F}_{\text{opt}}[k]) - \text{Im}(\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]) \right\|_F^2 \right\},\end{aligned}\quad (22)$$

where $\mathcal{L}_k \triangleq \|\mathbf{F}_{\text{opt}} - \mathbf{F}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F^2$ denotes the loss of the k th subcarrier.

By leveraging matrix calculus techniques [42], we can firstly derive the gradient of $\mathcal{L}_k$ with respect to the real and imaginary part of $\bar{\mathbf{F}}_{\text{RF}}$. Then we apply the chain rule to derive the gradient of $\mathcal{L}_k$ with respect to the quantized angle $\bar{\Phi}$ as

$$\begin{aligned}\nabla_{\bar{\Phi}} \mathcal{L}_k &= \{[\text{Re}(\mathbf{F}_{\text{opt}}[k]) - \text{Re}(\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k])]\text{Re}(\mathbf{F}_{\text{BB}}[k])^T\} \odot 2 \sin \bar{\Phi} \\ &\quad + \{[\text{Im}(\mathbf{F}_{\text{opt}}[k]) - \text{Im}(\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k])]\text{Im}(\mathbf{F}_{\text{BB}}[k])^T\} \odot 2 \sin \bar{\Phi} \\ &\quad + \{[\text{Re}(\mathbf{F}_{\text{opt}}[k]) - \text{Re}(\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k])]\text{Im}(\mathbf{F}_{\text{BB}}[k])^T\} \odot 2 \cos \bar{\Phi} \\ &\quad - \{[\text{Im}(\mathbf{F}_{\text{opt}}[k]) - \text{Im}(\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k])]\text{Re}(\mathbf{F}_{\text{BB}}[k])^T\} \odot 2 \cos \bar{\Phi}.\end{aligned}\quad (23)$$

Then, the gradient of $\mathcal{L}$ w.r.t. the unquantized phase shift Φ can be derived as

$$\nabla_{\Phi} \mathcal{L} = \frac{1}{K} \left(\sum_{k=0}^{K-1} \nabla_{\bar{\Phi}} \mathcal{L}_k \right) \odot \frac{\partial \bar{\Phi}}{\partial \Phi}. \quad (24)$$

The $\partial \bar{\Phi} / \partial \Phi$ term in (24) is clearly not differentiable at the quantization boundary, where the value of each entry is given as

$$\begin{aligned}\left[\frac{\partial \bar{\Phi}}{\partial \Phi} \right]_{m,n} &= \frac{\partial \lfloor [\Phi]_{m,n} / \Delta_{\phi} \rfloor \Delta_{\phi}}{\partial [\Phi]_{m,n}} \\ &= \begin{cases} \infty, & [\Phi]_{m,n} = \Delta_{\phi}(k + 0.5), k \in \mathbb{Z} \\ 0, & \text{otherwise} \end{cases}.\end{aligned}\quad (25)$$

The discontinuity in (25) will invalidate any gradient-based optimization approach as the gradient is either infinity or zero. To tackle this challenge, we utilize the DSQ function proposed in [35] to replace the ideal quantizer with the following hyperbolic functions,

$$\text{DSQ}_{\phi}(\phi, \kappa) = m_i + \frac{s \tanh(\kappa(\phi - m_i)) \Delta_{\phi}}{2}, \quad \phi \in \mathcal{R}_i, \quad (26)$$

where $\mathcal{R}_i = \{r \in \mathbb{R} \mid (i-1)\Delta_{\phi} < x \leq i\Delta_{\phi}\}$ partitions the space of real number into consecutive ranges of size Δ_{ϕ} , $m_i = (i-0.5)\Delta_{\phi}$ is the midpoint of each range, i can be any integer, $s = 1/\tanh(0.5\kappa\Delta_{\phi})$ is a scaling coefficient to guarantee smooth connection of DSQ function between two adjacent ranges, and κ is a tunable parameter to control the similarity between DSQ function and the ideal quantizer. The DSQ functions with different κ are depicted in Fig. 5. It is

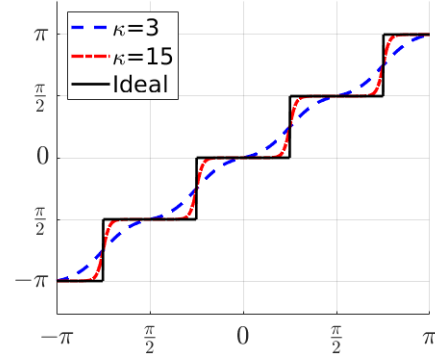


Fig. 5. Illustration of the differentiable soft quantization function when $\Delta_{\phi} = \pi/2$ with different value of κ , and comparison with the ideal quantizer.

worth noting that, unlike the DSQ function in [35], the value of the quantized phase shift is neither upper- nor lower-bounded since the exponential function in $\bar{\mathbf{F}}_{\text{RF}}$ calculation will inherently act as a wrapping function. This relaxation will facilitate the algorithm design, as will be shown later.

THz channel is said to be sparse since the number of multipaths is usually quite limited [36]. In such a scenario, the columns of the all-digital precoder $\mathbf{F}_{\text{opt}}$ very likely are highly correlated with the steering vectors aligned to the path angles [11]. Leveraging this structure, the proposed algorithm initializes Φ as the phase of steering vectors with pointed angles selected from a pre-defined set of angles $\mathcal{A}$ with coarse resolution, which is formulated as

$$\Phi = \angle([\mathbf{a}_t(\phi_1, \theta_1, f_c) \dots \mathbf{a}_t(\phi_{N_{\text{RF}}}, \theta_{N_{\text{RF}}}, f_c)]), \quad (27)$$

where $\{(\phi_i, \theta_i)\}_{i=1}^{N_{\text{RF}}}$ are selected from N_{RF} directions in $\mathcal{A}$ with the highest correlation with the channel matrix, and can be expressed as

$$\{(\phi_i, \theta_i)\}_{i=1}^{N_{\text{RF}}} = \arg \max_{\mathcal{A}' \subset \mathcal{A}, |\mathcal{A}'| = N_{\text{RF}}} \sum_{k=0}^{K-1} \|\mathbf{H}[k] \mathbf{a}(\phi, \theta, f_k)\|_F^2. \quad (28)$$

Afterward, the algorithm quantizes Φ to the predefined resolution Δ_{ϕ} and builds $\bar{\mathbf{F}}_{\text{RF}}$ accordingly. The baseband precoders $\mathbf{F}_{\text{BB}}[k]$ are then initialized with the least-squared solution as

$$\mathbf{F}_{\text{BB}}[k] = (\bar{\mathbf{F}}_{\text{RF}}^* \bar{\mathbf{F}}_{\text{RF}})^{-1} \bar{\mathbf{F}}_{\text{RF}}^* \mathbf{F}_{\text{opt}}[k]. \quad (29)$$

Next, we calculate $\mathcal{L}$ as (21) and approximate the gradient by replacing the non-differentiable term in (24) with the continuous and differentiable DSQ function as

$$\nabla_{\Phi} \mathcal{L} \approx \frac{1}{K} \left(\sum_{k=0}^{K-1} \nabla_{\bar{\Phi}} \mathcal{L}_k \right) \odot \frac{\partial \text{DSQ}_{\phi}(\Phi, \kappa^{(i)})}{\partial \Phi}, \quad (30)$$

where i denotes iteration index. We then use the approximate gradient to update the unquantized phase shift with a learning rate γ adapted from [43]. This process is repeated several times in each inner loop. We calculate $\mathbf{F}_{\text{BB}}$ with (29) at the beginning of each outer loop and increase the value of κ according to

$$\kappa^{(i)} = (\kappa_{\text{max}} - \kappa_{\text{start}})(1 - \beta^{i-1}) + \kappa_{\text{start}}, \quad (31)$$

Algorithm 1 Proposed Quantized-Phase Hybrid Beamforming Design Algorithm

Input: $\{\mathbf{F}_{\text{opt}}[k]\}_{k=0}^{K-1}$, Δ_ϕ

- 1: **Initialize:**
- 2: Initialize $\bar{\Phi}$ by (27), $\kappa^{(1)} = \kappa_{\text{start}}$ and $\gamma^{(1)} = \gamma^{\text{init}}$
- 3: $\bar{\Phi} = \left\lfloor \frac{\bar{\Phi}}{\Delta_\phi} \right\rfloor \cdot \Delta_\phi$
- 4: $\bar{\mathbf{F}}_{\text{RF}} = \exp(j\bar{\Phi})$
- 5: **for** $i = 1$ to N_{out} **do**
- 6: $\mathbf{F}_{\text{BB}}[k] = (\bar{\mathbf{F}}_{\text{RF}}^* \bar{\mathbf{F}}_{\text{RF}})^{-1} \bar{\mathbf{F}}_{\text{RF}}^* \mathbf{F}_{\text{opt}}[k]$
- 7: **for** $n_i = 1$ to N_{in} **do**
- 8: $\bar{\Phi} = \left\lfloor \frac{\bar{\Phi}}{\Delta_\phi} \right\rfloor \cdot \Delta_\phi$
- 9: $\bar{\mathbf{F}}_{\text{RF}} = \exp(j\bar{\Phi})$
- 10: $\mathcal{L} = \frac{1}{K} \sum_{k=0}^{K-1} \|\mathbf{F}_{\text{opt}}[k] - \bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F^2$
- 11: Approximate $\nabla_{\Phi} \mathcal{L}$ with (30)
- 12: $\bar{\Phi} = \bar{\Phi} - \gamma^{(n_i)} \cdot \nabla_{\Phi} \mathcal{L}$
- 13: Calculate $\gamma^{(n_i+1)}$ by [43]
- 14: **end for**
- 15: Calculate $\kappa^{(i+1)}$ by (31)
- 16: $\bar{\mathbf{F}}_{\text{RF}} = \exp(j\bar{\Phi})$
- 17: **end for**
- 18: $\mathbf{F}_{\text{BB}}[k] = \sqrt{N_s} \frac{\mathbf{F}_{\text{BB}}[k]}{\|\bar{\mathbf{F}}_{\text{RF}} \mathbf{F}_{\text{BB}}[k]\|_F}$, $k = 0, \dots, K-1$
- 19: **return** $\bar{\mathbf{F}}_{\text{RF}}, \{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1}$

where $0 < \beta < 1$ is a decay parameter, κ_{start} and κ_{max} are the initial and final values of κ . An increment in κ increases the resemblance between the hard and soft quantizers, forcing the phase shift at the quantization boundary to shift toward either side of the boundary as the slope increases. The pseudo-code of the proposed algorithm is listed in Algorithm 1, where γ^{init} denotes the initial learning rate and N_{in} and N_{out} denote the number of iterations in each inner loop and outer loop, respectively. From step 2 to step 4, the algorithm initializes parameters and pre-quantizes the phase shift. Phase shifts are then adjusted in the inner loop from step 7 to step 14. $\mathbf{F}_{\text{BB}}$ is derived with step 6 before the start of each inner loop. At the end of each outer loop, we adjusted the value of κ in step 15. Finally, in step 18, $\mathbf{F}_{\text{BB}}$ is scaled to meet the power constraint.

B. Limited-Resolution Hybrid Beamforming With TTD-Assisted Architecture

This subsection extends Algorithm 2 to the TTD-assisted hybrid beamforming architecture shown in Fig. 3(a). We first define the quantized time-delay matrix as follows.

$$\bar{\mathbf{T}} = \left\lfloor \frac{\text{clamp}(\mathbf{T}, \tau_{\text{max}}, \tau_{\text{min}})}{\Delta_\tau} \right\rfloor \cdot \Delta_\tau, \quad (32)$$

where $\bar{\mathbf{T}} \in \mathbb{R}^{N_{\text{TTD}} \times N_{\text{RF}}}$ is the quantized time-delay matrix. $\text{clamp}(\tau, \tau_{\text{max}}, \tau_{\text{min}}) \triangleq \max(\min(\tau, \tau_{\text{max}}), \tau_{\text{min}})$ is a clamping function to rectify the value of τ within $[\tau_{\text{min}}, \tau_{\text{max}}]$. τ_{max} , τ_{min} and Δ_τ are the maximum time delay, the minimum time delay and the resolution of TTDs,

respectively. Then, we formulate the loss function as

$$\begin{aligned} \mathcal{L} &= \frac{1}{K} \sum_{k=0}^{K-1} \mathcal{L}_k \\ &= \frac{1}{K} \sum_{k=0}^{K-1} \|\mathbf{F}_{\text{opt}}[k] - (\bar{\mathbf{F}}_{\text{PS}} \odot \bar{\mathbf{F}}_{\text{TTD}}[k]) \mathbf{F}_{\text{BB}}[k]\|_F^2, \end{aligned} \quad (33)$$

where $\bar{\mathbf{F}}_{\text{TTD}}[k]$ is the frequency-dependent phase shift of $\bar{\mathbf{T}}$ at the k th subcarrier, constructed using (12) with $\mathbf{T}$ replaced by $\bar{\mathbf{T}}$.

We then derive the gradient of $\mathcal{L}_k$ with respect to the quantized phase shift $\bar{\Phi}$ as the following element-wise expression

$$\begin{aligned} [\nabla_{\bar{\Phi}} \mathcal{L}_k]_{m,n} &= \frac{\partial \mathcal{L}}{\partial [\text{Re}(\bar{\mathbf{F}}_{\text{RF}})]_{m,n}} \left[-\sin(\bar{\phi}_{m,n}) \cos(2\pi f_k \bar{\tau}_{g(m),n}) \right. \\ &\quad \left. + \cos(\bar{\phi}_{m,n}) \sin(2\pi f_k \bar{\tau}_{g(m),n}) \right] \\ &\quad + \frac{\partial \mathcal{L}}{\partial [\text{Im}(\bar{\mathbf{F}}_{\text{RF}})]_{m,n}} \left[\cos(\bar{\phi}_{m,n}) \cos(2\pi f_k \bar{\tau}_{g(m),n}) \right. \\ &\quad \left. + \sin(\bar{\phi}_{m,n}) \sin(2\pi f_k \bar{\tau}_{g(m),n}) \right], \end{aligned} \quad (34)$$

where $\bar{\phi}_{m,n}$ denotes $[\bar{\Phi}]_{m,n}$, $\bar{\tau}_{\ell,n}$ denotes $[\bar{\mathbf{T}}]_{\ell,n}$ and $g: \{m \in \mathcal{P}_\ell\} \rightarrow \ell$ is a mapping function that maps PS index to the TTD index that the PS is connected. We can then approximate the gradient with the DSQ function by substituting (34) into (30). Next, we calculate the gradient of $\mathcal{L}_k$ with respect to the quantized time delay $\nabla_{\bar{\mathbf{T}}} \mathcal{L}_k$, whose entries are calculated as

$$\begin{aligned} [\nabla_{\bar{\mathbf{T}}} \mathcal{L}_k]_{\ell,n} &= \sum_{m \in \mathcal{P}_\ell} 2\pi f_k \left\{ \frac{\partial \mathcal{L}}{\partial [\text{Re}(\bar{\mathbf{F}}_{\text{RF}})]_{m,n}} \left[-\cos(\bar{\phi}_{m,n}) \sin(2\pi f_k \bar{\tau}_{\ell,n}) \right. \right. \\ &\quad \left. \left. + \sin(\bar{\phi}_{m,n}) \cos(2\pi f_k \bar{\tau}_{\ell,n}) \right] \right. \\ &\quad \left. + \frac{\partial \mathcal{L}}{\partial [\text{Im}(\bar{\mathbf{F}}_{\text{RF}})]_{m,n}} \left[-\sin(\bar{\phi}_{m,n}) \sin(2\pi f_k \bar{\tau}_{\ell,n}) \right. \right. \\ &\quad \left. \left. - \cos(\bar{\phi}_{m,n}) \cos(2\pi f_k \bar{\tau}_{\ell,n}) \right] \right\}. \end{aligned} \quad (35)$$

Analogous to (30), $\nabla_{\bar{\mathbf{T}}} \mathcal{L}$ can also be approximated by the DSQ function as

$$\nabla_{\bar{\mathbf{T}}} \mathcal{L} \approx \frac{1}{K} \left(\sum_{k=0}^{K-1} \nabla_{\bar{\mathbf{T}}} \mathcal{L}_k \right) \odot \frac{\partial \text{DSQ}_\tau(\mathbf{T}, \kappa^{(i)})}{\partial \mathbf{T}}, \quad (36)$$

where DSQ_τ is the soft quantization function for the time delay. Specifically, it is given by

$$\text{DSQ}_\tau(\tau, \kappa) = \begin{cases} \tau_{\text{max}}, & \tau \geq \tau_{\text{max}} \\ m_i + \frac{s \tanh(\kappa(\tau - m_i)) \Delta_\tau}{2}, & \tau \in \mathcal{R}_i \\ \tau_{\text{min}}, & \tau \leq \tau_{\text{min}} \end{cases}, \quad (37)$$

where $\mathcal{R}_i$ is the quantization interval defined as $\mathcal{R}_i = \{r \in \mathbb{R} \mid \tau_{\text{min}} + (i-1)\Delta_\tau < r \leq \tau_{\text{min}} + i\Delta_\tau\}$, for

Algorithm 2 Proposed Quantized-Phase-Delay Hybrid Beamforming Design Algorithm

Input: $\{\mathbf{F}_{\text{opt}}[k]\}_{k=0}^{K-1}$, Δ_ϕ , Δ_τ

- 1: **Initialize:**
- 2: Initialize $\bar{\Phi}$ with (27), $\mathbf{T}$ with (38)
- 3: $\kappa^{(1)} = \kappa_{\text{start}}$, $\gamma_\phi^{(1)} = \gamma_\phi^{\text{init}}$, and $\gamma_\tau^{(1)} = \gamma_\tau^{\text{init}}$
- 4: $\bar{\Phi} = \left\lfloor \frac{\Phi}{\Delta_\phi} \right\rfloor \cdot \Delta_\phi$
- 5: $\bar{\mathbf{T}} = \left\lfloor \frac{\text{clamp}(\mathbf{T}, \tau_{\text{max}}, \tau_{\text{min}})}{\Delta_\tau} \right\rfloor \cdot \Delta_\tau$
- 6: $\bar{\mathbf{F}}_{\text{RF}}[k] = \bar{\mathbf{F}}_{\text{PS}} \odot \bar{\mathbf{F}}_{\text{TTD}}[k]$
- 7: **for** $i = 1$ to N_{out} **do**
- 8: $\mathbf{F}_{\text{BB}}[k] = (\bar{\mathbf{F}}_{\text{RF}}^*[k] \bar{\mathbf{F}}_{\text{RF}}[k])^{-1} \bar{\mathbf{F}}_{\text{RF}}^*[k] \mathbf{F}_{\text{opt}}[k]$
- 9: **for** $n_i = 1$ to N_{in} **do**
- 10: $\bar{\Phi} = \left\lfloor \frac{\Phi}{\Delta_\phi} \right\rfloor \cdot \Delta_\phi$
- 11: $\bar{\mathbf{T}} = \left\lfloor \frac{\text{clamp}(\mathbf{T}, \tau_{\text{max}}, \tau_{\text{min}})}{\Delta_\tau} \right\rfloor \cdot \Delta_\tau$
- 12: $\bar{\mathbf{F}}_{\text{RF}}[k] = \bar{\mathbf{F}}_{\text{PS}} \odot \bar{\mathbf{F}}_{\text{TTD}}[k]$
- 13: $\mathcal{L} = \frac{1}{K} \sum_{k=0}^{K-1} \|\mathbf{F}_{\text{opt}}[k] - \bar{\mathbf{F}}_{\text{RF}}[k] \mathbf{F}_{\text{BB}}[k]\|_F^2$
- 14: Approximate $\nabla_{\Phi} \mathcal{L}$ by (30)
- 15: Approximate $\nabla_{\mathbf{T}} \mathcal{L}$ by (36)
- 16: $\Phi = \Phi - \gamma_\phi^{(n_i)} \cdot \nabla_{\Phi} \mathcal{L}$
- 17: $\mathbf{T} = \mathbf{T} - \gamma_\tau^{(n_i)} \cdot \nabla_{\mathbf{T}} \mathcal{L}$
- 18: Calculate $\gamma_\phi^{(n_i+1)}$ and $\gamma_\tau^{(n_i+1)}$ as in [43]
- 19: **end for**
- 20: Calculate $\kappa^{(i+1)}$ by (31)
- 21: $\bar{\mathbf{F}}_{\text{RF}}[k] = \bar{\mathbf{F}}_{\text{PS}} \odot \bar{\mathbf{F}}_{\text{TTD}}[k]$, $k = 0, \dots, K-1$
- 22: **end for**
- 23: $\mathbf{F}_{\text{BB}}[k] = \sqrt{N_s} \frac{\mathbf{F}_{\text{BB}}[k]}{\|\bar{\mathbf{F}}_{\text{RF}}[k] \mathbf{F}_{\text{BB}}[k]\|_F}$, $k = 0, \dots, K-1$
- 24: **return** $\{\bar{\mathbf{F}}_{\text{RF}}[k]\}_{k=0}^{K-1}$, $\{\mathbf{F}_{\text{BB}}[k]\}_{k=0}^{K-1}$

$i = 1 \dots \lfloor (\tau_{\text{max}} - \tau_{\text{min}}) / \Delta_\tau \rfloor$, and the middle point m_i is defined as $m_i = \tau_{\text{min}} + (i - 0.5) \Delta_\tau$. Note that $\text{DSQ}_\tau(\cdot, \cdot)$ is different from $\text{DSQ}_\phi(\cdot, \cdot)$ in (26) as we have to limit the values of time delay within the range $[\tau_{\text{min}}, \tau_{\text{max}}]$.

Given the approximations in (30) and (36), we formulate Algorithm 2 to derive the phase shift and time delay for the TTD-assisted beamformer with limited resolution. In Step 2, $\mathbf{T}$ is initialized with the equivalent time delay at f_c corresponding to the average phase shift of PSs within P_ℓ , given as

$$[\mathbf{T}]_{\ell, n} = \frac{1}{N_t / N_{\text{TTD}}} \sum_{m \in P_\ell} \frac{[\Phi]_{m, n}}{2\pi f_c}, \quad (38)$$

where Φ is initialized with (27). The phase shift and time delay are pre-quantized in step 4 and step 5 to construct the quantized RF precoder in step 6, and are updated in step 16 and step 17 of the inner loop by the DSQ approximation, and with adaptive learning rates $(\gamma_\phi, \gamma_\tau)$, where $\gamma_\phi^{\text{init}}$ and $\gamma_\tau^{\text{init}}$ denote the initial learning rates for the phase shift and the time delay, respectively. The outer loop calculates $\mathbf{F}_{\text{BB}}[k]$ in step 8 according to the updated $\bar{\mathbf{F}}_{\text{RF}}[k]$ from the inner loop, and update κ with (31) in step 20. Finally, $\mathbf{F}_{\text{BB}}[k]$ is scaled in step 23 to meet the power constraint.

V. SIMULATION RESULTS

This section presents the simulation results of the two proposed algorithms. The THz channel is constructed by (5), with channel parameters derived by the NYUSIM software [36], where the number of clusters N_{cl} follows a Poisson distribution and the number of rays L_i within each cluster i follows an exponential distribution. The time delay of each ray and each cluster is generated according to the well-known Saleh-Valenzuela model [44], where the delay difference between consecutive rays and clusters is exponentially distributed. The cluster AoD and AoA are uniformly chosen from $[0, 2\pi)$, whereas EoD and EoA are generated by Gaussian distributions with non-zero mean EoD and EoA. Finally, ray angular offsets are generated within each cluster using zero-mean Gaussian distributions with different angular spreads. The center frequency f_c is set to 140GHz in the sub-THz band with the number of subcarriers $K = 2048$ and a total transmission bandwidth of 4GHz. All the simulation results are averaged over 100 channel realizations.

A. Simulation Results of the PS-Only Hybrid Beamformer

This subsection demonstrates the numerical performance of the hybrid beamforming weights derived from Algorithm 1 for the PS-only architecture shown in Fig. 1. We consider a system with an 8×32 URA at the transmitter and a 4×4 URA at the receiver, where the array spacing is set to half wavelength $\lambda_c/2$ of the center frequency. $N_s = 4$ data streams are transmitted with $N_{\text{RF}} = 4$ RF chains at both the transmitter and the receiver. The parameters for Algorithm 2 are summarized as follows, $\gamma^{\text{init}} = 0.064$, $\kappa_{\text{start}} = 0.25$, $\kappa_{\text{max}} = 60$, $\beta = 0.99$, $N_{\text{in}} = 20$ and $N_{\text{out}} = 60$.

In Fig. 6, we plot the spectral efficiency versus SNR, where 3-bit PSs are assumed. B is the number of phase shift bits, and the SNR is defined as $\frac{P}{\sigma_n^2}$. The simulation result is compared with the performance of the all-digital optimal beamformer, the quantized hybrid beamforming algorithms given in [13], [16], and [20], and PE-AltMin [14] with infinite PS resolution. It can be seen that the proposed algorithm can achieve near-optimal performance with a PS resolution of only 3-bit, and the achievable spectral efficiency is the highest as compared to other hybrid beamforming algorithms.

In Fig. 7, we plot the spectral efficiencies of the proposed algorithm adopting different PS resolutions to demonstrate the effect of PS resolution on spectral efficiency. As shown in the figure, increasing the PS resolution enhances the spectral efficiency. In addition, the precoding performance of the proposed algorithm with 3-bit PS resolution is comparable to that of an all-digital precoder. Besides, the proposed algorithm also outperforms other precoding algorithms in all PS resolutions. Finally, with PS resolution greater or equal than 3, the proposed algorithm can outperform PE-AltMin [14] algorithm, which requires infinite resolution PS.

To validate the effectiveness of the proposed algorithm with different configurations of the hybrid beamformer, Fig. 8 compares the spectral efficiency of the proposed algorithm with other precoding schemes when a different number of RF chains are used in the system. In this case, we set the SNR to

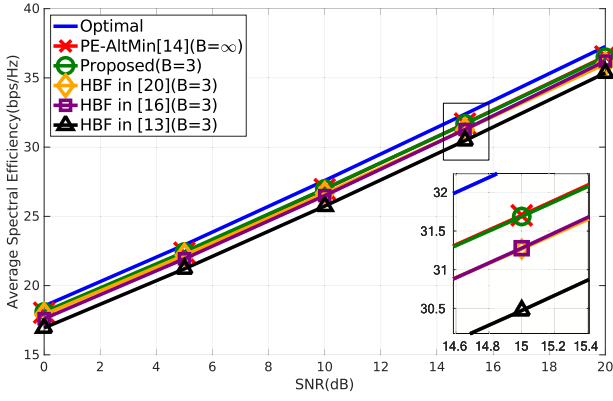


Fig. 6. Spectral efficiency versus SNR of different hybrid beamforming algorithms with 3-bit PS resolutions ($N_t = 8 \times 32$, $N_r = 4 \times 4$, $N_{RF} = 4$, $N_s = 4$).

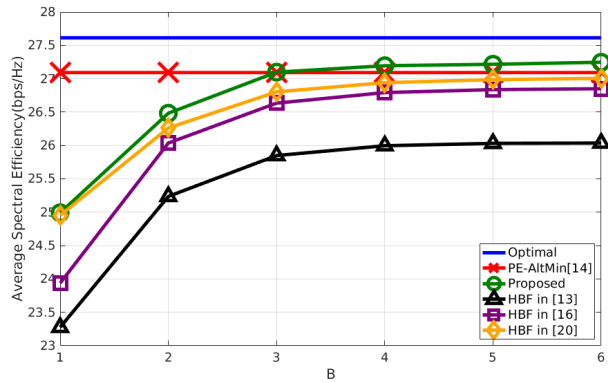


Fig. 7. Spectral efficiency versus PS resolution B of different hybrid beamforming algorithms with different PS resolutions at 10dB SNR ($N_t = 8 \times 32$, $N_r = 4 \times 4$, $N_{RF} = 4$, $N_s = 4$).

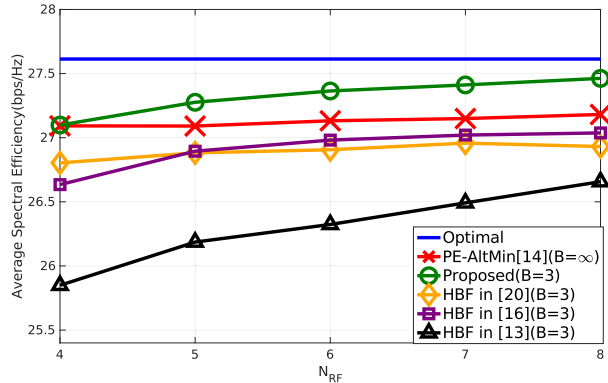


Fig. 8. Spectral efficiency versus N_{RF} of different hybrid beamforming algorithms at 10dB SNR with 3-bit PS resolution ($N_t = 8 \times 32$, $N_r = 4 \times 4$, $N_s = 4$).

10dB, and the PS resolution is set to 3-bit. It can be observed that, as N_{RF} increases, the spectral efficiency also increases because of the additional degree of freedom. Moreover, compared with other quantized precoding methods, the proposed algorithm can achieve the highest spectral efficiency among all N_{RF} . Finally, the performance of the proposed algorithm with 3-bit PS resolution also outperforms the infinite resolution PE-AltMin algorithm. In comparison, PE-AltMin [14] lags behind in spectral efficiency as N_{RF} increases.

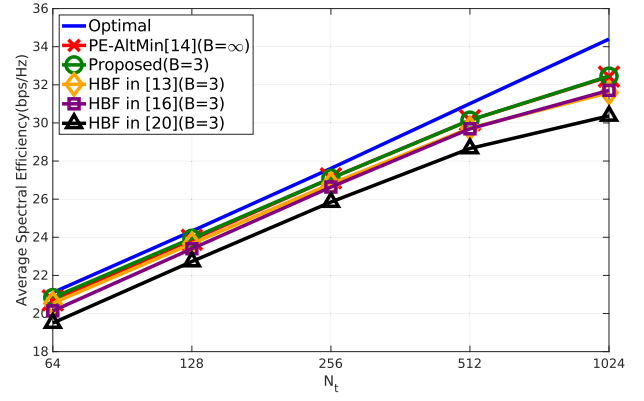


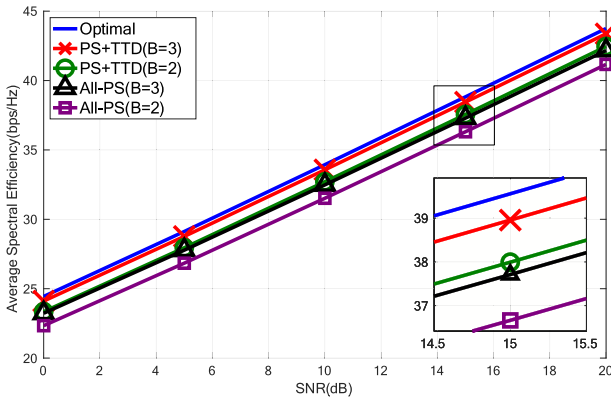
Fig. 9. Spectral efficiency versus number of transmit antennas N_t of different hybrid algorithms with different PS resolutions at 10dB SNR ($N_t = 8 \times 8, 8 \times 16, \dots, 8 \times 128$, $N_r = 4 \times 4$, $N_{RF} = 4$, $N_s = 4$).

Fig. 9 evaluates the spectral efficiency of the proposed algorithm when URAs of different apertures are used in the transmitter. The URA is assumed to have 8 antenna elements along the vertical axis, and we change the element number along the horizontal axis from 8 to 128 to adjust the total number of antennas of the URA from 64 to 1024. It can be seen from the figure that as the number of antennas grows, the spectral efficiency will also increase because of the higher beamforming gain. In addition, the proposed algorithm performs better than other methods with a 3-bit PS in all antenna sizes. However, the figure also shows that the spectral efficiency gap between the all-digital and hybrid beamformer increases as the number of antennas grows. This is the impact of beam squint, as we mentioned in Sec. II-C, which can be effectively mitigated with the help of TTDs.

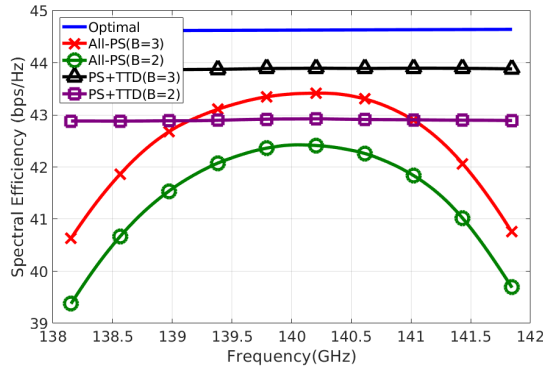
B. Simulation Results of the TTD-Assisted Hybrid Beamformer

The effectiveness of the proposed TTD-assisted hybrid beamforming algorithm is demonstrated in this subsection. We consider an 8×128 URA at the transmitter and a 4×4 URA at the receiver. When an antenna array with such a large-scale aperture is present, the beam squint effect becomes non-negligible and severely degrades the performance of the PS-only hybrid beamforming. In the following simulations, the numbers of data streams and RF chains are assumed to be 4. The parameters for Algorithm 2 are listed as follows, $\gamma_\phi^{\text{init}} = 0.064$, $\gamma_\tau^{\text{init}} = 0.001$, $\kappa_{\text{start}} = 0.25$, $\kappa_{\text{max}} = 60$, $\beta = 0.99$, $N_{\text{in}} = 20$ and $N_{\text{out}} = 80$.

Fig. 10 shows the spectral efficiency of the TTD-assisted beamformer with quantized phase shift and time delay values derived from Algorithm 2. Each TTD is assumed to be connected to a 4×8 PS sub-array with delay resolution Δ_τ set to $1/8$ the center frequency's period. We plot the results with 2-bit and 3-bit PS resolutions and compare them with the spectral efficiency of the PS-only beamformers with phase shift values obtained by Algorithm 2. As shown in Fig. 10(a), the average spectral efficiency of the TTD-assisted beamformer is higher than that of the PS-only beamformer in all SNR regions when the exact PS resolution is used. The higher spectral efficiency of the TTD-assisted beamformer is due



(a)



(b)

Fig. 10. Performance of TTD-assisted beamforming compared with PS-only beamforming ($N_t = 8 \times 128, N_{RF} = N_s = 4, V = 4, U = 8, N_{TTD} = 32, \Delta\tau = 1/8f_c$). (a) Spectral efficiency versus SNR with different PS resolutions. (b) Spectral efficiency versus frequency with different PS resolutions at 20dB SNR.

to the successful elimination of beam squint, as shown in Fig. 10(b), where we plot the average spectral efficiency of each subcarrier. The figure shows that the TTD-assisted beamformers have relatively constant spectral efficiencies across the signal band. In contrast, the PS-only beamformers suffer significant spectral efficiency degradation at band edges due to the beam squint effect.

Fig. 11 plots the minimum subcarrier spectral efficiency across the frequency bandwidth with a different number of transmit antennas to assess the degree of spectral efficiency degradation caused by beam squint. We consider a URA with 8 antenna elements along the vertical axis and 8 to 128 along the horizontal axis. Each TTD is assumed to be connected to a 4×4 PS subarrays with a resolution of $1/(8f_c)$. For comparison purposes, we also include the results of the PS-only beamformer designed with Algorithm 2 and other hybrid beamforming algorithms [13], [20] with a phase shift resolution of 3-bit. As illustrated in Fig. 11, the TTD-assisted beamformer does not suffer from severe SE decay as the number of antenna elements increases. In comparison, the spectral efficiency of the PS-only beamformer begins to deviate from that of the all-digital beamformer as the number of antennas grows. It is also worth noting that if the array aperture is not sufficiently long, such that only mild squinting occurs, TTD may not be required. As we can observe from Fig. 11, the spectral efficiency difference between the TTD-assisted

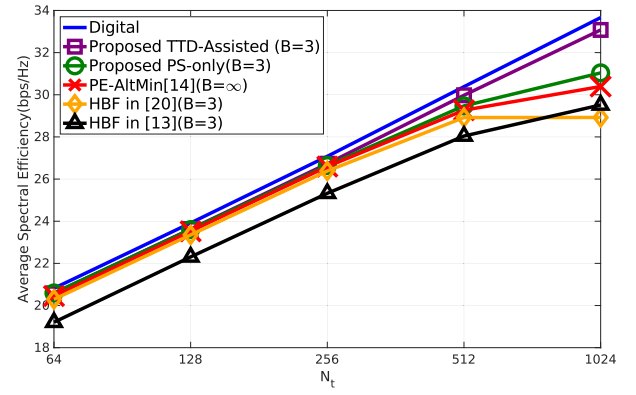


Fig. 11. Minimum spectral efficiency versus number of transmit antennas with TTD-assisted beamformer at 10dB SNR ($N_t = 8 \times 8$ to $8 \times 128, N_{RF} = N_s = 4, U = V = 4, \Delta\tau = 1/8f_c$).

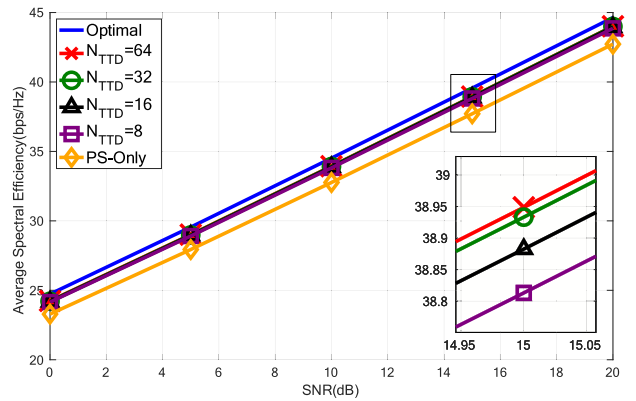


Fig. 12. Spectral efficiency versus SNR with different number of TTDs ($N_t = 8 \times 128, N_{RF} = N_s = 4, \Delta\tau = 1/8f_c, B = 3$).

beamformer and PS-only beamformer is very limited when the transmit URA has a small antenna number.

To study the effect of TTD number on spectral efficiency. In Fig. 12, we plot the spectral efficiency of TTD-assisted beamformer with different TTD number N_{TTD} at different SNR. We consider the following TTD connection schemes, $N_{TTD} = 8(1 \times 8), 16(1 \times 16), 32(2 \times 16), 64(2 \times 32)$, and assume the resolution of TTD is set to $1/(8f_c)$. While all the phase shifts of PSs are quantized to 3-bit. As a comparison, we also include the spectral efficiency of the PS-only beamformer. It can be observed from the figure that N_{TTD} can be significantly reduced to only 8 with no obvious spectral efficiency degradation while still yielding much higher performance than that of the PS-only beamformer. The result indicates that a small number of TTDs can sufficiently alleviate the beam squint effect, reducing the overall hardware complexity.

Fig. 13 depicts the change in spectral efficiency as we alter the TTD resolution. We plot the spectral efficiency of the TTD-assisted beamformer at different SNRs with different delay resolutions, where N_{TTD} is set to 32, each TTD is connected to a 4×8 PS subarray, and the resolution of PS is set to 3-bit. As expected, we can observe from Fig. 13 that increasing the delay interval of TTD reduces the overall spectral efficiency. However, we can also see that the spectral efficiency reduction is quite limited, indicating that fine-grain

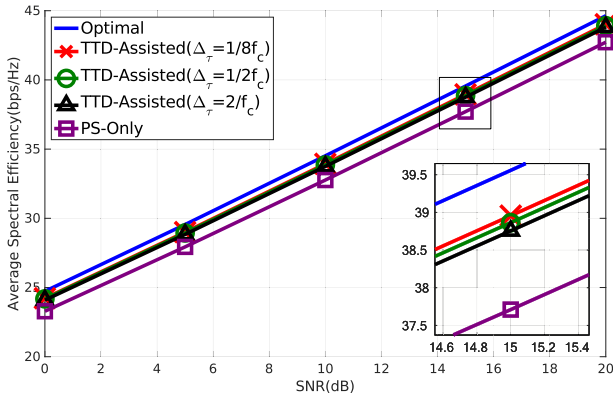


Fig. 13. Spectral efficiency versus SNR with different TTD resolution ($N_t = 8 \times 128$, $N_{RF} = N_s = 4$, $U = 4$, $V = 8$).

TTDs with a resolution equaling a fraction of the half-wavelength delay [27] may not be necessary to achieve high spectral efficiency. TTD with a coarse delay resolution is sufficient to reduce the beam squint effect and provide high, squint-free beamforming performance.

VI. CONCLUSION

This paper considers a wideband massive-MIMO hybrid beamforming system with a 3D sub-THz channel model. We proposed two algorithms based on gradient optimization techniques with differentiable soft quantization for efficiently designing hybrid beamforming coefficients with consideration of the limited resolution of precoding hardware. The first algorithm is for PS-only architecture, while the second is for TTD-assisted architecture. Simulation results show that the proposed algorithm can achieve a near-optimal spectral efficiency with a PS resolution of only 3-bit, which outperforms other state-of-the-art quantized hybrid beamforming algorithms. In addition, the proposed algorithm for TTD-assisted hybrid beamforming can effectively alleviate the beam squint effect and achieve near-optimal performance with only a small number of low-resolution TTDs.

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